

sports/tier1_sim_config.py

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1  """Generative defaults for Tier 1 / **540** soft-assignment + score/select.
2
3  =====
4  FOR LATER CHARLES - this is the KNOB FILE (like 530 CELL 2)
5  =====
6  Imported by: tier1_pool_assignment.AssignmentParams, Pass A/B bundles,
7               tier1_generative_eda.SelectionConfig.
8  Daily runners: sports/scripts/hero_model_reset_bundle.py
9                  sports/scripts/540_rho_ablation_bundle.py
10 Plain English: 3-Master_Plan/re_entry/04_Pass_A_and_Pass_B_in_Plain_English.md
11 Engine decode: sports/tier1_pool_assignment.py (READ-FIRST GLOSSARY there too).
12
13 Do not open tier1_cell10_playground_run.py for re-entry checklist work.
14
15 -----
16 READ-FIRST GLOSSARY - every symbol below is defined HERE before knobs use it.
17 Do not skip this block. Later constants assume you already know these words.
18 -----
19
20 WHAT THIS FILE IS
21   A list of default numbers/strings for one synthetic league draw + selection.
22   Changing a constant here changes the *default* Pass A/B / notebook run
23   unless a script overrides that field with replace() / state JSON.
24
25 THREE STEPS (same as tier1_pool_assignment)
26   (1) ASSIGN - seat fake players on teams.
27   (2) SCORE - ranking number  $S_i$  for each player.
28   (3) SELECT - pick winners (usually top  $K$  by  $S_i$ ).
29
30 CORE LETTERS
31    $A_i$  - ability of player  $i$  (drawn talent).
32    $T_j$  - team target for team  $j$  (soft assignment aims here).
33   pool - team / roster group (pool_id on each player).
34    $S_i$  - selection score (ranking weight only).
35    $K$  - how many get selected (=  $N_{SELECTED}$  below).
36    $Y_{selected}$  - 0/1 after SELECT.
37
38 POOL L (enters SCORE)
39   LOO - leave-one-out (exclude self when measuring teammates).
40    $L_Q$  - mean teammate ability (column poolq_loo). Same units as  $A$ .
41    $L_C$  - crowding among viable peers, on  $[0,1]$  (hard share or smooth).
42    $\theta$  - viability cutline ( $VIABILITY\_THETA$ ):  $A > \theta \Rightarrow$  "viable peer."
43    $\gamma$  - sharpness of soft viability ( $VIABILITY\_SHARPNESS$ ).
44
45 SCORE (Pass A toggles these)
46    $w$  / LOO_GAP_WEIGHT - weight on  $L$  in  $S = A - w \cdot L$ . Same role as Alex  $\lambda$ .
47    $\lambda$  (lambda) - Alex's name for that weight; in code,  $w \approx \lambda$ .
48   SELECTION_SCORE_MODE - "ability" ( $S=A$ ) or "loo_gap_plus_ability" ( $S=A-w \cdot L$ ).
49   LOO_POOL_L_MODE - which  $L$ : "quality" / "crowding" / "crowding_smooth".
50   CROWDING_L_Z_SCALE - same idea as  $l_{term\_scale}$  in the engine:
51                        $l$  = pool  $L$ ; term =  $L$ -piece of  $S=A-w \cdot L$ ; scale = units.
52                       None = auto ( $p_{90}(A)-p_{10}(A)$ ) when  $A$  is z-scored.
53
54 ASSIGN (Pass B toggles these)
55    $q$  / ASSIGNMENT_RHO - soft assortativity.  $0 \approx$  mix; larger  $\approx$  sit near  $T_j$ .
56                       NOT the same as  $\lambda/w$  (those are SCORE weights).
57    $\sigma$  / ASSIGNMENT_SIGMA - length-scale in the soft kernel ( $\sim 0.65$ ).
58    $\tau$  / ASSIGNMENT_TEMPERATURE - legacy twin of  $\sigma$ ; prefer  $q$  in 540 work.
59   soft vs sort-and-chop - soft = overlapping windows; sort_chop = hard slices.
60
61 SELECT
62   WINNER_SELECTION "C" - top  $K$  by score (Pass A/B default).
63   "A"/"B" - older stochastic rules.
64
65 TRACKS / PRESETS BELOW (legacy CELL 10 buttons - optional)
66   Track A POOL_530_* - match 530 empirical pool forensics (overlap / SD).
67   Track B MATCH_539_* - mimic 539 hard-assignment feel ( $\tau$  from calibration).
68   Layer 2 SELECTION_539_* - 539 teaching score (beta  $A$ , crowding_smooth,  $w=\lambda$ ).
69   For daily 540 re-entry, focus on the MAIN knobs ( $N\_TEAMS$  ... WINNER_SELECTION
70   and CROWDING_L_Z_SCALE). Preset blocks are for older playground buttons.
71
72 SECTION MAP
73   1. League size ( $N\_TEAMS$ , ROSTER_SIZE)
74   2. Team targets  $T_j$ 
75   3. Soft assignment ( $q$ ,  $\sigma$ ,  $\tau$ , preferential)
76   4. Ability draw  $A_i$ 

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77 5. Calibration targets (530 forensics numbers – for plots, not science knobs)
78 6. Selection / inverted-U replay (L mode, w, K, winner rule)
79 7. CELL 10 presets (530 / 539 tracks) – optional
80 8. Notebook run gates / plot flags
81 =====
82 " "
83
84 # =====
85 # 1. LEAGUE SIZE
86 # =====
87 # Empirical panel analysis in 538 still uses PipelineConfig (CELL 1) from
88 # sports_pipeline.config – do not duplicate those knobs here.
89
90 # N_TEAMS (J): how many synthetic teams (pools) in the fake league.
91 # PD11: J must be >> number of ventile / Plot B bins so bins are not empty.
92 N_TEAMS = 1000
93 # 538/530 ventiles often 8–20; keep N_TEAMS several times larger than K bins.
94
95 # N_TEAMS_SLIDER_MAX: CELL 10 widget cap only (not a hard algorithmic limit).
96 N_TEAMS_SLIDER_MAX = 3000
97
98 # ROSTER_SIZE: how many players sit on every team (fixed equal rosters).
99 ROSTER_SIZE = 15
100
101 # N_INDIVIDUALS: total fake players = teams × roster size (derived, do not edit alone).
102 N_INDIVIDUALS = N_TEAMS * ROSTER_SIZE
103
104
105 # =====
106 # 2. TEAM TARGETS T_j
107 # =====
108 # T_j = one number per team; soft assignment tries to seat players with A ≈ T.
109
110 # TARGET_MEAN_DIST: which family draws T_j.
111 # "uniform" | "normal_clipped" | "empirical_530"
112 # empirical_530 = draw from scipy fit saved by 530 CELL 5b (same law as empirical A_i)
113 TARGET_MEAN_DIST = "uniform"
114
115 # TARGET_MEAN_LOW / HIGH: uniform range, or clip bounds for normal_clipped.
116 TARGET_MEAN_LOW = -0.5
117 TARGET_MEAN_HIGH = 0.5
118
119 # TARGET_MEAN_MU / SIGMA: used only when TARGET_MEAN_DIST == "normal_clipped".
120 TARGET_MEAN_MU = 0.0
121 TARGET_MEAN_SIGMA = 0.35
122
123
124 # =====
125 # 3. SOFT ASSIGNMENT (Pass B knobs:  $\rho$ ,  $\sigma$ )
126 # =====
127 # Soft  $\pi_{ij}$ : probabilistic seating near T_j (overlapping talent windows).
128
129 # ASSIGNMENT_KERNEL: shape of soft-match weights. "gaussian" | "cauchy"
130 ASSIGNMENT_KERNEL = "gaussian"
131
132 # ASSIGNMENT_RHO ( $\rho$ ): assignment assortativity – Pass B's main knob.
133 # 0 = max mixing (flat among open seats);  $\rho \uparrow$  = sharper match to T_j.
134 # NOT  $\lambda$  / LOO_GAP_WEIGHT (those live in SCORE).
135 ASSIGNMENT_RHO = 1.0
136
137 # ASSIGNMENT_SIGMA ( $\sigma$ ): fixed ability–target length-scale in the soft kernel.
138 # ( $\rho=1$ ,  $\sigma=0.65 \equiv$  legacy  $\tau=0.65$  on the old parameterization.)
139 ASSIGNMENT_SIGMA = 0.65
140
141 # ASSIGNMENT_TEMPERATURE ( $\tau$ ): legacy alias; keep equal to ASSIGNMENT_SIGMA.
142 # Old notebooks /  $\tau$  prose. Prefer  $\rho + \sigma$  in 540 work.
143 ASSIGNMENT_TEMPERATURE = 0.65
144
145 # USE_PREFERENTIAL_ATTACHMENT: if True, multiply soft weights by  $(n_j + k)^\alpha$ 
146 # (rich-get-richer on seats already filling). Off for 540 defaults.
147 USE_PREFERENTIAL_ATTACHMENT = False
148 # PREFERENTIAL_ALPHA ( $\alpha$ ): exponent when preferential attachment is on.
149 PREFERENTIAL_ALPHA = 0.35
150 # PREFERENTIAL_K ( $k$ ): offset in  $(n_j + k)^\alpha$  so empty teams are not weight 0.
151 PREFERENTIAL_K = 1.0
152
153
154 # =====
155 # 4. ABILITY DRAW A_i
156 # =====
157 # A_i = latent talent for each synthetic player.
158

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159 # ABILITY_DRAW: which distribution invents A_i.
160 # "uniform_01" | "beta_2_2" (539 Alex) | "normal_clipped" | "normal_plus_student_t"
161 # "empirical_530" = draw from scipy fit saved by 530 CELL 5b (within-season z perf)
162 # Scale note: normal_clipped is z-ish → crowding L needs CROWDING_L_Z_SCALE / auto.
163 ABILITY_DRAW = "normal_clipped"
164
165 # ABILITY_MEAN / SD: location and scale for normal_* draws.
166 ABILITY_MEAN = 0.0
167 ABILITY_SD = 1.0
168
169 # ABILITY_CLIP_LOW / HIGH: hard floor/ceiling after normal_* draws.
170 ABILITY_CLIP_LOW = -2.5
171 ABILITY_CLIP_HIGH = 3.5
172
173 # ABILITY_STUDENT_T_*: heavy-tail add-on when ABILITY_DRAW includes student-t noise.
174 # DF = degrees of freedom; SCALE = 0 means that noise is off.
175 ABILITY_STUDENT_T_DF = 4
176 ABILITY_STUDENT_T_SCALE = 0.25
177
178
179 # =====
180 # 5. CALIBRATION TARGETS (from 530 forensics – for validation plots)
181 # =====
182 # These are “what good soft assignment should look like,” not Pass A/B science knobs.
183 # Within-season z, PPM-based panel, ~2011–2021, APPLY_ANALYSIS_FILTERS True.
184
185 CALIBRATION_NOTE = (
186     "Match 530 CELL 8: coverage >> 1 at center; CELL 5: median roster SD ~0.8; "
187     "CELL 6: weak positive mean–SD correlation, not forced negative."
188 )
189 # EMPIRIC_MEDIAN_ROSTER_SD_Z: typical within-team ability SD on z scale (~0.8).
190 EMPIRIC_MEDIAN_ROSTER_SD_Z = 0.80
191 # EMPIRIC_COVERAGE_PEAK_ORDER: order-of-magnitude overlap peak (re-read 530 after refresh).
192 EMPIRIC_COVERAGE_PEAK_ORDER = 3000
193
194
195 # =====
196 # 6. SELECTION / INVERTED-U REPLAY (Pass A knobs: score mode, L, w, K)
197 # =====
198 # Selection = draft pick, tenure, promotion, etc. Bins are on LOO pool L (not teams).
199
200 # LOO_POOL_L_MODE: which L enters the SCORE.
201 # "quality" → L_Q (poolq_loo)
202 # "crowding" → L_C hard share (pool_c_loo)
203 # "crowding_smooth" → LOO mean  $\sigma(\gamma(A-\theta))$  (pool_c_smooth_loo; 539 / Pass A congestion)
204 LOO_POOL_L_MODE = "quality"
205
206 # VIABILITY_THETA ( $\theta$ ): cutline for “viable peer” when building L_C.
207 # Default ≈ 530 CELL 5d median drafted within-season z perf (PPM, ever-drafted).
208 VIABILITY_THETA = 0.7546158731868137
209
210 # VIABILITY_SHARPNESS ( $\gamma$ ): logistic sharpness in soft viability  $\sigma(\gamma(A-\theta))$ .
211 # Higher  $\gamma$  → closer to a hard threshold at  $\theta$  (539 simulate_viable_peer_congestion).
212 VIABILITY_SHARPNESS = 18.0
213
214 # GENERATIVE_N_BINS: how many bins on the Plot B / inverted-U X-axis.
215 GENERATIVE_N_BINS = 20
216
217 # GENERATIVE_POOLQ_BINNING: how to cut those bins. "quantile" | "equal_width"
218 GENERATIVE_POOLQ_BINNING = "quantile"
219
220 # N_SELECTED (K): how many players get Y_selected=1 (top K under winner "C").
221 N_SELECTED = 1500
222
223 # SELECTION_SCORE_MODE: how S_i is built.
224 # "ability" → S = A (talent-only /  $\lambda=0$ )
225 # "loo_gap_plus_ability" → S = A - w·L (quality: gap vs L_Q; crowding: vs L_C)
226 SELECTION_SCORE_MODE = "loo_gap_plus_ability"
227
228 # LOO_GAP_WEIGHT (w): weight on the L-term in S = A - w·L. Same role as Alex  $\lambda$ .
229 # Crowding mode reuses this knob as the “crowding weight.”
230 LOO_GAP_WEIGHT = 0.5
231
232 # WINNER_SELECTION: how scores become winners.
233 # "A" = weighted sample without replacement
234 # "B" = Bernoulli
235 # "C" = top-K deterministic ← Pass A/B default
236 WINNER_SELECTION = "C"
237
238 # CROWDING_L_Z_SCALE: unit-matching multiplier for crowding L (same as l_term_scale).
239 # Name decode: crowding = L_C mode; L = pool L; Z_SCALE = scale when A is z-scored.
240 # When using z-scored ability + crowding L, assign_selection multiplies L_C by this

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241 # (or auto p90(A)-p10(A) when None). Pin a positive float (e.g. 4.0) to freeze it.
242 CROWDING_L_Z_SCALE = None
243
244
245 # =====
246 # 7. CELL 10 PRESETS (two calibration tracks – do not mix casually)
247 # =====
248 #
249 # Track A – POOL_530_* ("530 pool cal" button): match real MBB forensics (530 CELLS 5–8).
250 #  $\tau \approx 0.65 \rightarrow$  median roster SD  $\sim 0.8$ , coverage peak  $\gg 1$ . NOT the 539 Alex assignment story.
251 #
252 # Track B – MATCH_539_* ("539 assign cal" button): mimic 539 sort-chop assignment (CELL 10b/10c)
253 # before congestion in the selection score.  $\tau$  from auto_calibrate_tau; re-run 10c after
254 # changing ability_draw, J, or seed. Soft assign cannot fully match hard sort-chop – 10b
255 # will still show a gap on roster SD / coverage.
256 #
257 # Layer 0 (both tracks): ability-only top-K (w=0). Layer 2+: turn on crowding_smooth + w.
258 #
259 # Layer 2 – 539 teaching selection (CELL 10 "539 selection" button):
260 # beta_2_2 A on [0,1], T_j uniform [0,1], crowding_smooth, w =  $\lambda$  from reference JSON.
261 # Keeps 530 pool assignment ( $\tau \approx 0.65$ ) unless you switch to 539 assign cal separately.
262
263 # --- Track A – 530 empirical pool calibration -----
264 POOL_530_ASSIGNMENT_TEMPERATURE = 0.65
265 POOL_530_ABILITY_DRAW = "normal_clipped"
266 POOL_530_TARGET_MEAN_DIST = "uniform"
267 POOL_530_PREFERENTIAL_ALPHA = 0.0
268 POOL_530_SELECTION_SCORE_MODE = "ability"
269 POOL_530_LOO_GAP_WEIGHT = 0.0
270 POOL_530_LOO_POOL_L_MODE = "quality"
271 POOL_530_WINNER_SELECTION = "C"
272
273 # Deprecated aliases (530 track) – old names; same values as POOL_530_* above.
274 MINIMAL_ABILITY_DRAW = POOL_530_ABILITY_DRAW
275 MINIMAL_TARGET_MEAN_DIST = POOL_530_TARGET_MEAN_DIST
276 MINIMAL_PREFERENTIAL_ALPHA = POOL_530_PREFERENTIAL_ALPHA
277 MINIMAL_SELECTION_SCORE_MODE = POOL_530_SELECTION_SCORE_MODE
278 MINIMAL_LOO_GAP_WEIGHT = POOL_530_LOO_GAP_WEIGHT
279 MINIMAL_LOO_POOL_L_MODE = POOL_530_LOO_POOL_L_MODE
280 MINIMAL_WINNER_SELECTION = POOL_530_WINNER_SELECTION
281
282 # --- Track B – 539 assignment mimicry (Layer 0) -----
283 #  $\tau$  from CELL 10c at J=1000, seed=42, rho=0.88
284 MATCH_539_ASSIGNMENT_TEMPERATURE = 0.108
285 MATCH_539_N_TEAMS = 1000
286 MATCH_539_ROSTER_SIZE = 15
287 MATCH_539_RANDOM_SEED = 42
288 MATCH_539_ABILITY_DRAW = "normal_clipped"
289 MATCH_539_TARGET_MEAN_DIST = "uniform"
290 MATCH_539_TARGET_MEAN_LOW = -0.5
291 MATCH_539_TARGET_MEAN_HIGH = 0.5
292 MATCH_539_PREFERENTIAL_ALPHA = 0.0
293 MATCH_539_N_SELECTED = 1500
294 MATCH_539_SELECTION_SCORE_MODE = "ability"
295 MATCH_539_LOO_GAP_WEIGHT = 0.0
296 MATCH_539_LOO_POOL_L_MODE = "quality"
297 MATCH_539_WINNER_SELECTION = "C"
298 MATCH_539_VIABILITY_THETA = 0.72
299 MATCH_539_VIABILITY_SHARPNESS = 10.0
300 # Full-scale 539 notebook reference (tier1_539_reference_settings.json): N=30_000, J=2000, seed=1,
301 # ability=beta_2_2 on [0,1], T_j uniform [0,1], n_selected=3000 (10%). Re-run 10c after switching.
302 # Sweep hint (J=1000): beta_2_2  $\rightarrow \tau \approx 0.116$  (seed 42) or  $\tau \approx 0.149$  (seed 1); 538 SD  $\sim 0.21$  vs 539  $\sim 0.11$ .
303 MATCH_539_FULL_N_TEAMS = 2000
304 MATCH_539_FULL_ROSTER_SIZE = 15
305 MATCH_539_FULL_RANDOM_SEED = 1
306 MATCH_539_FULL_ABILITY_DRAW = "beta_2_2"
307 MATCH_539_FULL_TARGET_MEAN_LOW = 0.0
308 MATCH_539_FULL_TARGET_MEAN_HIGH = 1.0
309 MATCH_539_FULL_N_SELECTED = 3000
310
311 # --- Layer 2 – 539 selection score on 530-calibrated pools -----
312 # Commensurate [0,1] A and L_C (beta draw + crowding_smooth +  $\lambda$  from reference JSON).
313 SELECTION_539_ABILITY_DRAW = "beta_2_2"
314 SELECTION_539_TARGET_MEAN_LOW = 0.0
315 SELECTION_539_TARGET_MEAN_HIGH = 1.0
316 SELECTION_539_LOO_POOL_L_MODE = "crowding_smooth"
317 SELECTION_539_LOO_GAP_WEIGHT = 0.55 #  $\lambda$  in tier1_539_reference_settings.json
318 SELECTION_539_SELECTION_SCORE_MODE = "loo_gap_plus_ability"
319 SELECTION_539_VIABILITY_THETA = 0.72
320 SELECTION_539_VIABILITY_SHARPNESS = 10.0
321 SELECTION_539_WINNER_SELECTION = "C"
322

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323 # Deprecated aliases (rename only – same values as main N_SELECTED / score mode).
324 N_PROMOTED = N_SELECTED
325 PROMOTION_SCORE_MODE = SELECTION_SCORE_MODE
326
327
328 # =====
329 # 8. NOTEBOOK RUN GATES / PLOT FLAGS (legacy CELL 10 playground)
330 # =====
331 # These do not change Pass A/B bundle science; they gate optional notebook cells.
332
333 # RUN_GENERATIVE_ASSIGNMENT_DEMO: if True, 538-era notebook runs assignment demo cells.
334 RUN_GENERATIVE_ASSIGNMENT_DEMO = False
335 # RUN_REPLAY_530_STYLE_CHECKS: if True, run 530-style overlap / SD replay checks.
336 RUN_REPLAY_530_STYLE_CHECKS = False
337
338 # RANDOM_SEED: default RNG seed when a caller does not pass its own Generator.
339 RANDOM_SEED = 42
340
341 # SHOW_PLOT_A: CELL 10 playground – interval-overlap Plot A (530 CELL 8 analog).
342 SHOW_PLOT_A = True
343 # SHOW_PLOT_B: CELL 10 playground – inverted-U Plot B (selection rate vs LOO L bins).
344 SHOW_PLOT_B = True
345 # SHOW_PLOT_B_TEAM_MEAN: False → Plot B x = L_Q (530 comparability);
346 #   True → x = pool team_mean (539 notebook style).
347 SHOW_PLOT_B_TEAM_MEAN = False
348 # SHOW_PLOT_C: CELL 10 playground – ability histogram Plot C (530 CELL 5b overlay).
349 SHOW_PLOT_C = True
```